

Using humic acid for nutrient load reduction and red tide mitigation

An integrated field experiment and artificial intelligence approach from lysimeter to watershed scale

! PROBLEM

- Agricultural runoff is the dominant source of TN and TP in South Florida watersheds
- Sandy soils → rapid fertilizer leaching to groundwater & surface waters
- Nutrient enrichment drives harmful algal blooms (e.g., *Karenia brevis*) in the Gulf of America



💡 HYPOTHESIS

Humic acid (HA) applied with organic amendments will:

- 1 Enable ~50% reduction in fertilizer application by enhancing nutrient retention in sandy soils
- 2 Significantly reduce TN and TP leaching to groundwater and surface water, lowering downstream nutrient loads



🎯 GOAL

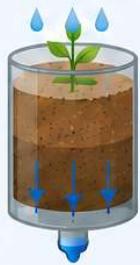
Reduce nutrient loads and mitigate red tide through humic acid application, validated across multiple scales using field data and watershed and AI modeling.



INTEGRATED MULTI-SCALE APPROACH

1. LYSIMETER SCALE

Controlled soil column experiments



- Quantify reduction in TN and TP leaching
- Test 50% N & P fertilizer reduction

2. FIELD SCALE

On-farm demonstration (Immokalee, FL)



- Validate 50% fertilizer reduction potential
- Quantify improvement in farm economy
- Develop AI-powered dashboard

3. WATERSHED SCALE

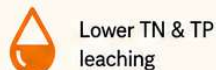
AI + hydrologic modeling



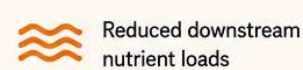
- Watershed model integration
- Estimate reduction in downstream loads and red tide probability using machine learning



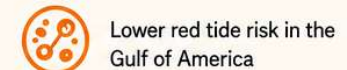
50% reduction in fertilizer use



Lower TN & TP leaching



Reduced downstream nutrient loads

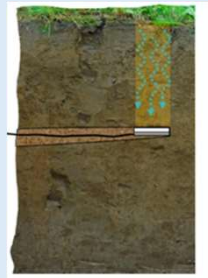


Lower red tide risk in the Gulf of America

Humic Acid for reducing nutrient loads and HAB risks

Task 1: Lysimeter-Scale Experiment

Quantify Nutrient Leaching to Demonstrate Improvement in Groundwater – Surface Water Quality



Suction plate lysimeter (ecoTech)

Task 2 Field-Scale Validation

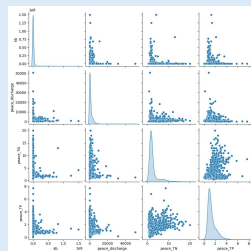
Quantify Crop Yield and Quality to Demonstrate Improvement in Farm Economy while Improving Water Quality



Tomato field, UF-IFAS, Immokalee

Task 3: Watershed-Scale Modeling

Quantify Gulf Nutrient Load Reduction and Demonstrate Decrease in Red Tide Risk

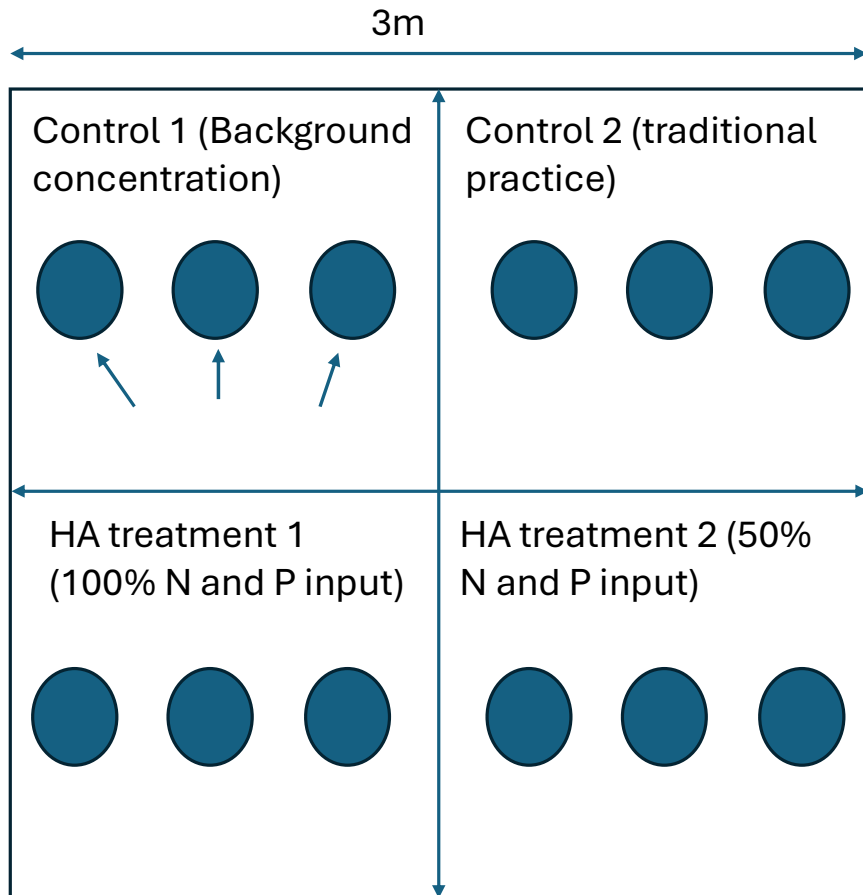


Higher nutrient loads a Peace River watershed correspond to increased bloom intensity

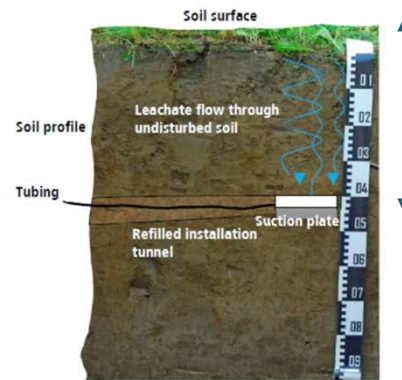
Technological Innovations

- (A) Humic Acid to Minimize Nutrient Leaching and Improve Farm Economy
- (B) UAV to Document Crop Health and Support Stakeholder Engagement
- (C) Machine Learning Model to Predict Red Tide Mitigation given Agricultural Best Management Practices with Humic Acid
- (D) AI-LLM Powered Decision-Support Dashboard for Agricultural Stakeholders

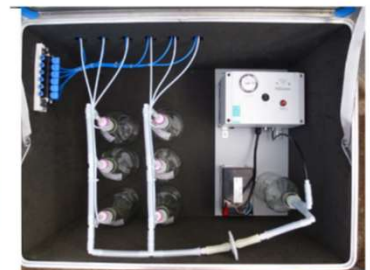
Task 1: Demonstrate at lysimeter scale the role humic acid in reducing leaching and improving water quality



Find root depth and stay below root zone 5 cm – 10 cm



Suction plates

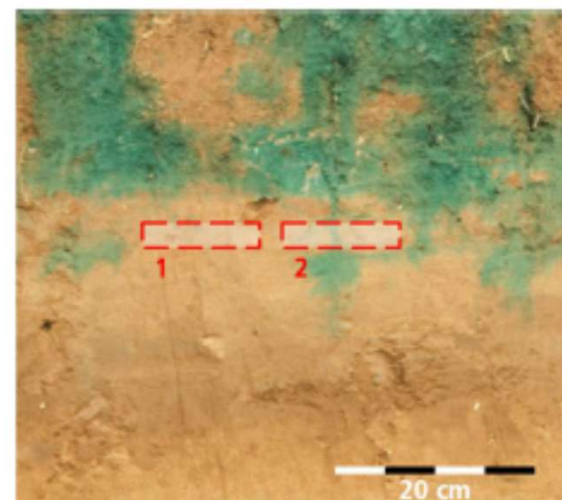
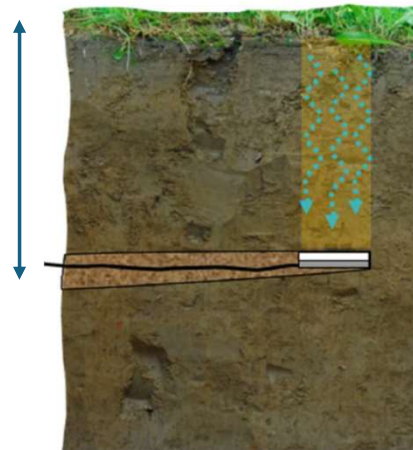


Sampling station

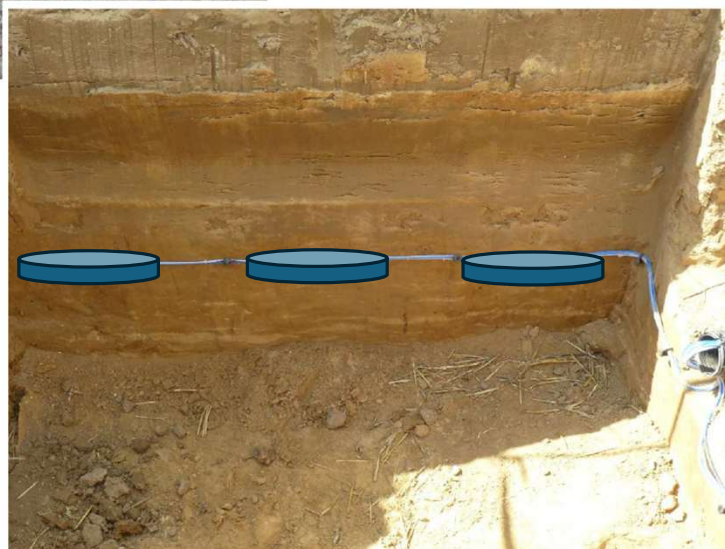
Lysimeter experiment to quantify TN and TP leaching to groundwater (3 replicates)



Find root depth and stay below
root zone 5 cm – 10 cm

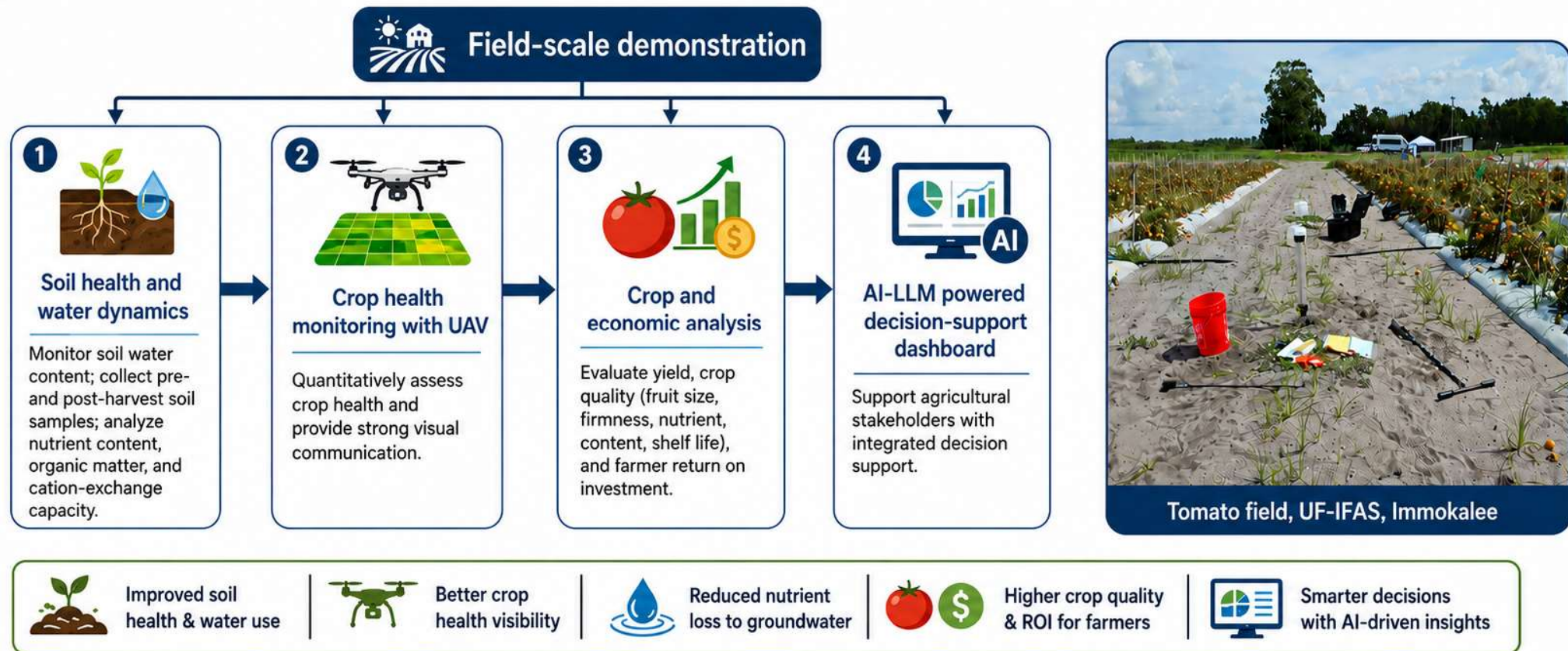


3 replicates needed



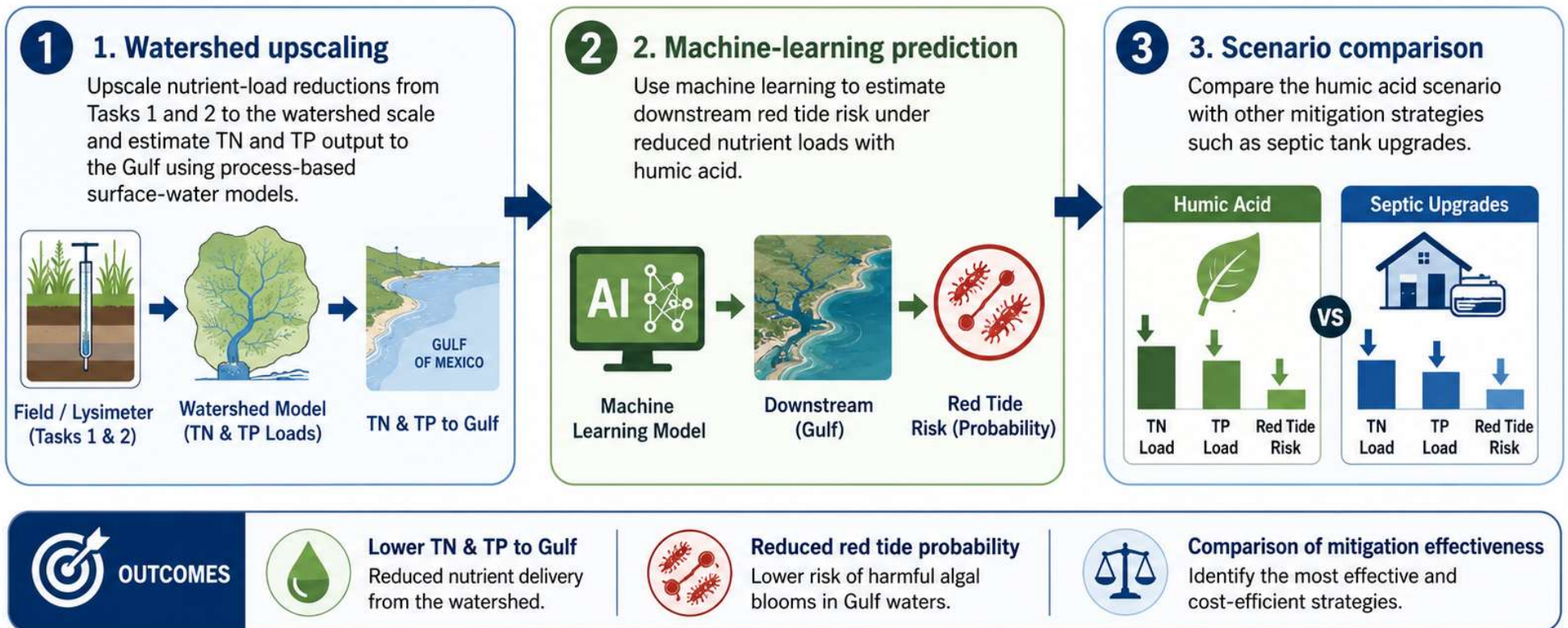
Task 2 Demonstrate at field scale the efficacy of humic acid in improving crop quality and reducing cost

Standard practice (100% nutrients) vs HA treatment (50% reduced nutrient input) under commercial agricultural conditions.



Task 3 Demonstrate at watershed scale the efficacy of HA in reducing nutrient input to the Gulf water and decreasing of the probability of red tide occurrence

Watershed-scale upscaling, AI prediction, and scenario comparison



Publication plan

Task 1 – Lysimeter Experiment (Water Research)

Focus: Mechanistic understanding of nutrient leaching reduction

Title: Humic acid reduces nitrogen and phosphorus leaching from sandy soils: Evidence from controlled lysimeter experiments

Contribution: process-level quantification

Emphasize:

- TN/TP mass flux reduction
- Soil retention mechanisms (CEC, water holding)
- This paper establishes causality at the soil scale

Task 2 – Field Validation (JoEM, JoCP, or AWM)

Focus: Field validation and economic feasibility

Title: Field-scale validation of humic acid as a best management practice for reducing fertilizer use and improving water quality and farm profitability

Contribution: real-world applicability

Emphasize:

- 50% fertilizer reduction feasibility
- Yield, quality, and ROI
- Environmental–economic tradeoffs
- This paper establishes adoptability and scalability at farm level

Task 3a - Watershed Modeling (JoH, or JoH-RS)

Focus: Upscaling to watershed nutrient load reduction

Title: Upscaling agricultural nutrient management: Watershed-scale impacts of humic acid adoption on nitrogen and phosphorus loading

Contribution: linking field results to watershed outcomes

Emphasize:

- Scenario analysis (baseline vs HA adoption)
- Load reductions to receiving waters
- Sensitivity and uncertainty
- This paper establishes regional hydrologic impact

4) Task 3b - ML Red Tide Prediction (ES&T or STOTEN)

Focus: Linking nutrient reduction to HAB risk

Title: Reducing harmful algal bloom risk through agricultural nutrient management: A machine learning assessment of red tide response to watershed nutrient scenarios

Contribution: predictive environmental response

Emphasize:

- ML model performance
- Nutrient–HAB relationship
- Scenario-based risk reduction
- This paper establishes coastal/ecological relevance

Publication plan

Final Synthesis (ES&T, STOTEN, or JoCP)

Focus: Integrated farm-to-coast framework

Title: From farm management to coastal resilience: A multiscale framework linking agricultural nutrient reduction to harmful algal bloom mitigation

Contribution: conceptual and integrative advance

Emphasize:

- Soil → field → watershed → coastal system linkage
- Quantified HAB risk reduction
- Scalable environmental solution
- This paper establishes generalizable framework

Narrative flow: How the papers fit together ? Each paper answers a distinct question:

- Task 1 Mechanism (WR): Does it work mechanistically?
- Task 2 Real-world validation (JoEM): Does it work in practice and economically?
- Task 3a Watershed scaling (JoH): Does it matter at watershed scale?
- Task 3b Ecological impact - HABs (STOTEN): Does it reduce environmental risk - HABs?
- Final Integrated framework (ES&T): Can this transform how we manage nutrient pollution?